INCH-POUND

MIL-A-82785(OS)
5 February 1990
SUPERSEDING
WS 22368A
9 December 1986

MILITARY SPECIFICATION

ADHESIVE, TWO PART, HIGH STRENGTH

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of two part epoxy adhesive, referred to herein as the adhesive.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8040

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MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1002 Strength Properties of Adhesives in Shear by Tension Loading (DoD adopted)

ASTM D 1876 Peel Resistance of Adhesives (T-Peel Test) (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

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(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3.) in accordance with 4.3.

3.2 Material. The adhesive shall be a two-part adhesive resin consisting of an epoxy resin (Part A) and an amine-type curing agent (Part B).

3.2.1 Adhesive formulation. The adhesive formulation shall conform to the following:

a. Part A 100 ± 1 parts by weight

b. Part B 22 ± 1 parts by weight

NOTE: The adhesive must be mixed thoroughly in order to obtain the performance specified herein.

3.2.2 Kits. The adhesive shall be supplied in kits consisting of separate containers for part A and part B.

3.3 Chemical and physical properties. The chemical and physical properties shall conform to table I.

3.4 Safety.

3.4.1 Toxic products and formulations. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.4.2 Material Safety Data Sheet (MSDS). The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

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TABLE I. Chemical and physical properties.

Property	Mixed Material	
	Minimum	Maximum
Peel strength, T-peel, pounds per inch of width (piw)	30	____
Lap shear strength, pounds per square inch (psi)	4000	____
Water resistance, 24 hours at 25 degrees Celsius (°C) followed by peel strength, piw	25	____
Pot life, minutes	30	____

3.5 **Shelf life.** The adhesive components shall meet the requirements of this specification for a period of 12 months from the date of the quality conformance inspection (see 4.4) when packaged as specified in section 5 and stored at 80° Fahrenheit (F) maximum.

3.6 **Workmanship.** The adhesive component materials shall be free from agglomeration, foreign matter, separation, dilution, or contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Toxicological product formulations. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection (see 3.1) shall consist of the examination and tests specified in 4.4. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of examinations and tests specified in table II.

TABLE II. Quality conformance inspection.

Examination and Tests	Requirement	Test Method
Visual examination	3.2, 3.6, and section 5	4.5.1
Peel strength	3.3	4.5.2
Lap shear strength	3.3	4.5.3
Water resistance	3.3	4.5.4
Pot life	3.3	4.5.5
Packaging inspection	Section 5	4.5.6

4.4.1 Sampling.

4.4.1.1 Sampling for visual examination. Sampling for visual examination to determine conformance to 5.1 and 5.2 shall consist of a random selection of at least 10 percent of the lot. Sampling for visual examination to determine conformance to 3.6 shall consist of the sample selected in accordance with 4.4.1.2.

4.4.1.2 Sampling for tests. Sampling for tests shall consist of a minimum of at least 1 quart of part A with a sufficient quantity of part B for proper composition, with each part selected at random from each lot.

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4.4.2 **Lot.** A lot shall consist of all the epoxy resin and curing agent packaged separately and subject to inspection at one time. Each of the two lot components shall have been compounded and mixed in a single container during one mixing period without any change in the process or materials.

4.5 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Visual examination of the samples selected in 4.4.1.1 shall be conducted to determine conformance to 3.4, 5.1, and 5.2. Contractor certification verifying conformance to 3.2 and 3.2.2 shall be required (see 6.2). Conformance to 3.6 shall be determined by visual examination.

4.5.2 **Peel strength.** Peel strength shall be determined by a T-peel test in accordance with 4.5.2.2. The test panels shall be 6-inch by 12-inch panels of 2024-T3 aluminum alloy.

4.5.2.1 **Aluminum substrate preparation.** The aluminum panels shall be cleaned in a solvent or alkaline solution and subsequently etched with a solution consisting of the following:

a. 30.0 ± 1.5 parts deionized water

b. 10.0 ± 0.5 parts sulfuric acid (specific gravity 1.84)

c. 1.00 ± 0.05 part sodium dichromate

The etching solution temperature shall be 65 ± 3°C. The panels shall be immersed in the solution for 10 to 15 minutes, deionized water rinsed, air dried at room temperature for 15 minutes minimum, and then forced air dried for 20 minutes minimum in a 65 ± 10°C mechanical convection oven in which air passes the panels at a velocity of 100 to 200 feet per minute. The panels then shall be removed from the oven, cooled to room temperature, and used within 2 hours.

4.5.2.2 **Test procedure.** The adhesive shall be mixed in accordance with 3.2.1, and applied to a 6.0 ± 0.1 by 9.0 ± 0.1 inch area of the test panels which have been prepared in accordance with 4.5.2.1. The ASTM D 1876 test specimens shall be cured in an oven at 65 ± 3°C for 60 to 90 minutes and stored for a minimum of 20 hours at 23 ± 2°C before testing in accordance with ASTM D 1876 to determine peel strength.

4.5.3 **Lap shear strength.** Lap shear strength (aluminum to aluminum) shall be determined at room temperature in accordance with ASTM D 1002. The aluminum substrates shall be prepared in accordance with 4.5.2.1, and the adhesive shall

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be mixed in accordance with 3.2.1. The test specimens shall be cured in an oven at 65 ± 3°C for 60 to 90 minutes and stored for a minimum of 20 hours at 23 ± 2°C before testing.

4.5.4 **Water resistance.** Three specimens shall be prepared as specified in 4.5.2, except that prior to testing in accordance with ASTM D 1876 the following shall be performed:

a. Completely immerse each specimen in water for 24 ± 2 hours at room temperature (23 ± 2°C).

b. After immersion, the specimens shall be lightly wiped and air dried for 30 to 60 minutes at room temperature. The specimens shall then be tested for peel strength in accordance with ASTM D 1876.

4.5.5 **Pot life.** The adhesive shall be mixed in accordance with 3.2.1 and allowed to set for 30 ± 2 minutes. At the end of this time, peel specimens shall be prepared and tested in accordance with 4.5.2.

4.5.6 **Packaging inspection.** The packaging and marking of the adhesive compound shall be inspected to verify conformance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The adhesive compound shall be packaged to afford adequate protection against loss, contamination, deterioration and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.5. All containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178 or to other carrier rules and regulations as applicable to he mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

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f. Shelf life and inspection/test date for extension (see 3.5). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION:

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 6 MONTH PERIOD PROVIDED THE ADHESIVE COMPOUND, UPON RETEST, CONFORMS TO THE LAP SHEAR STRENGTH AND POT LIFE REQUIREMENTS SPECIFIED IN MIL-A-82785."

"TEMPERATURE SENSITIVE MATERIAL

DO NOT EXPOSE TO TEMPERATURE HIGHER THAN 80°F."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The adhesive is intended for general use in structural bonding of Standard Missile MK-104 Dual Thrust Rocket Motor components.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Whether first article is required (see 3.1 and 6.3).

d. Applicable contract provisions for Material Safety Data Sheet (see 3.4.2 and 6.4).

e. DOD FAR 52.246-15 (see 4.5.1).

f. Packaging required, if other than as specified in 5.1.1.

g. Special marking, if other than as specified in 5.2.2.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quality of the adhesive compound sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the

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right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Material Safety Data Sheets (MSDS). MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), SUBPART 23.3. Contracting officers will identify those activities requiring copies of completed MSDSs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Suggested source of supply. The following material has been found to meet the requirements of this specification: EA9309.2 NA, manufactured by HYSOL Division, The Dexter Corporation, Pittsburg, CA 94565. This listing is given for information only and is not restrictive.

6.6 Subject term (key word) listing.

Rocket Motor, Dual Thrust Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 8040-N157)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-82785	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
ADHESIVE, TWO-PART, HIGH STRENGTH

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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